Unit 2 Pre-Test

Scientific Notation

Standards assessed: NC.8.EE.3, NC.8.EE.4

Name: _____ **Date:** _____ **Period:** _____

SCORE 3.0 Basic Understanding

NC.8.EE.3

• I can write numbers in scientific notation and in standard form, like 45,000 = 4.5 × 10⁴.

Match each vocabulary term to its definition, fill in the parts of a number in scientific notation, then write each number as directed.

– scientific notation	**A.** A number written the ordinary way. *Example: 2,700 or 0.05*
– coefficient	**B.** The decimal factor in a × 10ⁿ — at least 1 and less than 10. *Example: the 8.3 in 8.3 × 10⁶*
– standard form	**C.** Shows a small number between 0 and 1 (move the decimal left). *Example: the −2 in 5 × 10⁻² = 0.05*
– base	**D.** A number written as a × 10ⁿ, where 1 ≤ a < 10. *Example: 8.3 × 10⁶*
– positive exponent	**E.** In a power of ten, the number that is multiplied repeatedly. *Example: the 10 in 10⁴ = 10 · 10 · 10 · 10*
– negative exponent	**F.** Shows a large number, 10 or greater (move the decimal right). *Example: the 6 in 8.3 × 10⁶ = 8,300,000*

Parts of a number in scientific notation — fill in the blanks

$$3.4 \times 10^{5}$$

_____ the 3.4 _____ the 10 _____ the 5

A positive exponent makes a **large** *number; a negative exponent makes a* **small** *one.*

1. *Write in scientific notation*

45,000

Answer: _____

2. *Write in standard form*

3×10^{4}

Answer: _____

3. *Write in scientific notation*

0.0062

Answer: _____

4. *Write in standard form*

7.1×10^{-2}

Answer: _____

5. *Write in scientific notation*

580,000

Answer: _____

6. *Fix the notation*

Is 12.5×10^{6} correct? Write the correct form.

Answer: _____

SCORE 4.0 Target Content

NC.8.EE.4

- *I can multiply and divide in scientific notation and say how many times bigger one number is.*

Multiply, divide, or compare. Write each answer in correct scientific notation.

7. *Multiply*

$(3 \times 10^4)(2 \times 10^3)$

Answer: _____

8. *Divide*

$(8 \times 10^9) \div (4 \times 10^5)$

Answer: _____

9. *Compare*

A blue whale has a mass of about 1.5×10^5 kg. A dog has a mass of about 3×10^1 kg. How many times greater is the whale's mass?

Answer: _____

10. *Compare*

How many times larger is 6×10^8 than 2×10^3?

Answer: _____

11. *Divide*

$(9 \times 10^{-2}) \div (3 \times 10^4)$

Answer: _____

12. *Multiply*

$(2.5 \times 10^5)(2 \times 10^{-8})$

Answer: _____

SCORE **5.0** **Advanced Content**

NC.8.EE.4

• I can multiply and divide with negative exponents and solve word problems in scientific notation.

Solve. Show your work. Write each answer in correct scientific notation (the first number must be at least 1 and less than 10).

13. *Multiply, then fix the form*

$(4 \times 10^{-3})(6 \times 10^{-5})$

Answer: _____

14. *Divide, then fix the form*

$(1.2 \times 10^{-4}) \div (4 \times 10^{-9})$

Answer: _____

15. *Multi-step, then fix the form*

$$\frac{(7.5 \times 10^{-6})(4 \times 10^{3})}{2.5 \times 10^{-4}}$$

Answer: _____

16. *Apply*

A red blood cell is about 8×10^{-6} meter wide. A grain of sand is about 6.4×10^{-4} meter wide. How many times wider is the grain of sand than the red blood cell? Write your answer in scientific notation.

Answer: _____

17. *Apply*

One bacterium has a mass of about 2×10^{-12} gram. A sample has 6×10^{8} bacteria. What is the total mass of the sample, in grams?

Answer: _____

18. *Apply*

A hard drive holds 2×10^{12} bytes. One song file uses about 4×10^{6} bytes. How many song files can the hard drive hold?

Answer: _____